

Weekly Progress Report

Amity School of Engineering & Technology

Project Group ID - 265

**Project Area – Data Science
(Machine Learning)**

B. Tech (8CSE-DS-1Y) VII Semester

WPR No. 7

Duration: 19/01/2026 to 25/01/2026

Students Weekly Progress Report (WPR) for Even Semester of session 2025-2026

To be filled by Students	
Students Name	1. Sagar Singh Bisht 2. Aakarsh Pab
Roll no.	1. 056 2. 050
Enrollment no.	1. A023167022056 2. A023167022050
Project Title finalized, if Yes, give name, if NO, give reason	Machine Learning based Predictive Model for Livability using Environmental and Socioeconomic indicators
Synopsis submitted	YES
Literature review	YES
Technical & Economical Feasibility	YES
Bill of Material	NA
Project Progress Schedule (PERT Chart)	NA
Design of critical components	-
Fabrication work (give %)	-
Experimental work (give %)	-
Result and Analysis	-
Report writing	-
Signature of students	

Work done in this week:

Worked on the web page design of the project.

Integrated streamlit web for the task and made a prototype modal of the web page ,how will it look

And also worked on the rent data for the prediction by entering it manually in the master dataset

Plan for Next Week

- Framework Initialization: Setup the Streamlit environment and define the application architecture to ensure the backend script can be imported as a module.
- UI/UX Design: * Construct a Sidebar Control Panel featuring st.slider widgets for each priority (Rent, Air Quality, Traffic, Amenities).
 - Implement Constraint Inputs (e.g., a numeric input for "Maximum Budget") to act as a hard filter before the similarity ranking begins.
- Dynamic Result Display: Create a "Results Gallery" using st.columns and st.metric to showcase the top 5 recommended neighborhoods with their respective similarity scores.

2. Integration & Real-Time Logic

- Backend-to-Frontend Bridge: Connect the Cosine Similarity function to the Streamlit session state so that the recommendations update dynamically as the user moves the sliders.
- Data Visualization: * Integrate st.map to provide a spatial view of the recommended areas.
 - Develop a Radar Chart (using Plotly) for each recommendation to show how well the neighborhood matches the user's specific priorities (e.g., "High match on Rent, low match on Metro")

To be filled with a Guide (strike off whichever is not applicable)	
The performance of students is satisfactory	
The performance of students is unsatisfactory	
A warning to be issued to student(s) (Name)	
Student was not well (Name)	
Date 25/01/2026	Signature of Guide